Online Continual Learning-Based Intrusion Detection in Industrial Control Networks

Iwan Donal Paska Manurung
Dept. of Electrical Engineering
Fakultas Teknik Universitas Indonesia
Depok, Indonesia
iwan.donal@ui.ac.id

Ruki Harwahyu
Dept. of Electrical Engineering
Fakultas Teknik Universitas Indonesia
Depok, Indonesia
ruki.h@ui.ac.id

***Abstract*—Digital transformation has rendered critical infrastructure and its supporting industrial control systems increasingly complex through the integration of advanced communication and computing capabilities. Consequently, Operational Technology (OT) networks have become progressively converged with Information Technology (IT) networks. While this integration significantly enhances process management and automation, it simultaneously increases the exposure of industrial control components to a diverse array of cyber threats. To address these emerging security challenges, the ability to identify specific threat vectors constitutes a primary defensive measure. Online Continual Learning-based Intrusion Detection Systems (IDS) offer high efficacy in detecting cyber threats, including novel attack patterns. This research addresses these challenges by implementing Online Continual Learning methods within industrial control networks. The proposed IDS is designed with a multi-layer architecture to maintain data confidentiality of industrial control processes. This study demonstrated that localized IDS models can achieve high reliability without compromising data privacy by exchanging telemetry data to central servers.**

***Index Terms*—intrusion detection, cybersecurity, industrial control system, online continual learning, resilient infrastructure, privacy-preserving security**

I. Introduction

The Industrial Control System (ICS) refers to an automated control mechanisms designed to manage, regulate, and automate manufacturing processes. Historically, ICS were designed as isolated systems, disconnected from public network connectivity such as internet. However, the emerging of digital transformation has fundamentally shifted this landscape from a closed architecture toward a more open, integrated model that bridges OT with IT. These integrated model introduce significant security vulnerabilities that threaten Confidentiality, Integrity, and Availability (CIA) [1].

To ensure effective mitigation, an IDSs were designed to identify active threats or unauthorized intrusions [1], typically operates in real-time, a capability defined by minimizing the latency between the occurrence of a threat and its subsequent detection. IDSs operate by collecting data across various layers and processing it centrally at the Enterprise layer of the Purdue Model for ICS [2]. While effective, this architecture poses risks to data privacy. Furthermore, ICS components often face constraints in terms of computation, memory, power supply, and communication. Such limitations present significant challenges for IDS deployment.

This research proposes an intrusion detection mechanism designed for localized threat identification, thereby eliminating the necessity of offloading raw data to external systems beyond ICS network. Under this localized architecture, the IDS minimizes network overhead by bypassing data center transmissions, communicating only high-level alerts upon detecting potential anomalies. To preserve data privacy, the framework adopts a multi-layered approach tailored to the ICS data hierarchy, specifically targeting the Edge and Fog layers. Both IDS tiers are engineered for localized data ingestion and processing; by maintaining the entire data lifecycle within the OT perimeter, the system achieves low-latency, real-time performance—a critical requirement facilitated by removing dependencies on the IT/OT communication bridge. Furthermore, the proposed IDS is optimized for computational efficiency, ensuring robust deployment on resource-constrained devices typical of legacy and modern industrial environments.

II. Practical Limitations In Industrial Control Systems

The integration of Machine Learning (ML) and Deep Learning (DL) has emerged as a prominent trend in Industrial Control System threat detection [3]. These methodologies enable dynamic data learning, facilitating the identification of novel patterns such as zero-day exploits. In IT environments, ML models typically utilize centralized architectures where audit data is aggregated and analyzed within a Cloud or Centralized ecosystem [5]. While cloud computing provides the high-performance resources necessary for managing heterogeneous data, complex models, and rapid inference, this centralized paradigm often conflicts with the inherent constraints of ICS, such as stringent low-latency requirements, intermittent connectivity, and resource-constrained environments.

Furthermore, the robustness of ML models is increasingly threatened by adversarial attacks [4]. In these scenarios, compromised data samples are intentionally introduced into the audit stream to deceive the model, leading to misclassification or failure to detect anomalies. Such vulnerabilities are particularly acute in cloud-based detection systems, where audit data may undergo unauthorized alteration during transmission, effectively transforming the data itself into an attack vector [5].

Data confidentiality presents an additional critical challenge for cloud-centric frameworks. Within this framework, the continuous transmission of telemetry to a central server is unavoidable [6], thereby introducing significant risks of data leakage and expanding the overall attack surface of the industrial infrastructure.

While offline machine learning-based IDSs offer reliable performance, their development requirements are not always applicable to the context of ICS. Components within ICS networks typically face constraints in computation, memory, and communication. Computational limitations prevent the IDS from processing large-scale data simultaneously, while memory constraints hinder the extensive data collection required by offline machine learning. Furthermore, regarding communication requirements, low-latency connectivity is a fundamental necessity to ensure the IDS can operate in real-time.

III. Online Continual Learning-Based Intrusion Detection Systems

Online Continual Learning (OCL) represents a system that possess the adaptive capability to evolve over changing operational environments by processing data stream instances sequentially. The widespread adoption of ML and Artificial Intelligence (AI) across industrial sectors necessitates models that offer the flexibility to navigate dynamic and fluctuating environments.

Traditional research predominantly utilizes static learning, where algorithms are trained on a static, stationary, and comprehensive dataset to generate a model for evaluating subsequent test data. However, in real-world industrial settings, both normal and anomalous data characteristics may shift over time—a phenomenon where statistical properties such as mean, variance, or autocorrelation evolve over time. To mitigate the limitations of static modeling, Online Continual Learning utilizes an incremental approach, specifically designed to facilitate real-time learning from continuous data streams [7]. This ensures that the system remains robust and relevant in the face of evolving environmental dynamics. Given that data accumulation can reach infinity, the model must efficiently process data streams that expand indefinitely over time within constrained resource environments.

The OCL adopts the prequential (test-then-train) and single-pass-data evaluation, where each single incoming data instance from the stream is first utilized for model testing—assessing the system's predictive accuracy on unseen data—prior to being incorporated into the model training phase. Despite its adaptability, OCL is inherently susceptible to catastrophic forgetting, a significant phenomenon in model training where the acquisition of new information results in the abrupt loss of past learned knowledge. This issue is frequently exacerbated by concept drift [8], where a shift in the underlying data distribution or statistical properties causes the model to prioritize current patterns at the expense of historical learning. Consequently, maintaining a balance between stability (retaining old knowledge) and plasticity (learning novel concepts) remains a primary challenge in developing robust OCL-based classification.

IV. Proposed Methodology

The design of the IDS in this study adopts the Purdue Model for Industrial Control Systems, where each architectural tier is equipped with a dedicated IDS module tailored to the specific data characteristics of specific layer. The experimental framework evaluates two distinct modules: the Edge-IDS and the Fog-IDS as in Fig. 1. Both systems operate independently, utilizing heterogeneous datasets determined by the specific data availability at their respective hierarchical levels.

The proposed IDS operates by utilizing an ensemble methodology comprising three distinct OCL models. This ensemble approach is utilized as an efficient strategy to enhance detection efficacy, leveraging the collective strengths of diverse algorithms to mitigate individual model biases [17]. The final classification is determined through a majority voting mechanism from the combined outputs. By integrating multiple predictive perspectives, this architecture ensures higher robustness and reliability, particularly when identifying complex and evolving attack patterns in high-stakes industrial environments.

The proposed design adopts a multi-layered IDS architecture that establishes a functional decoupling between the Edge and Fog layers. The Edge-IDS operates exclusively on localized data within the Edge perimeter, while the Fog-IDS is restricted to data ingested at the Fog level. This experimental design is specifically engineered to preserve data confidentiality by ensuring each IDS utilizes only the information available at its respective tier, the system ensures no cross-layer data exchange.

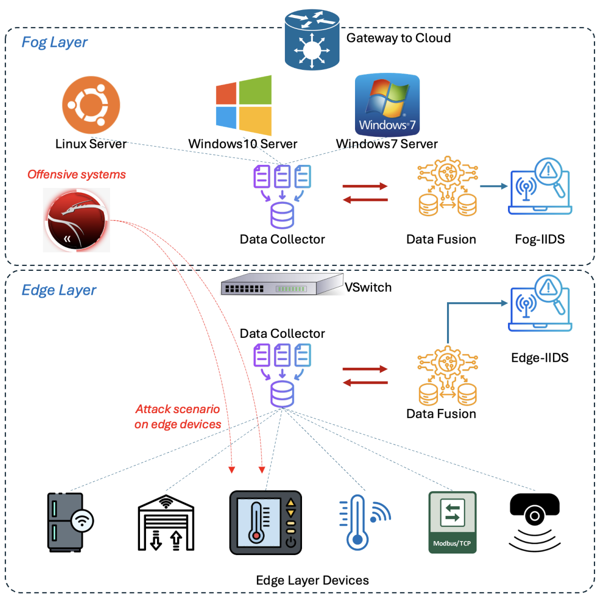

Fig. 1. Multi-layer architecture IDS.

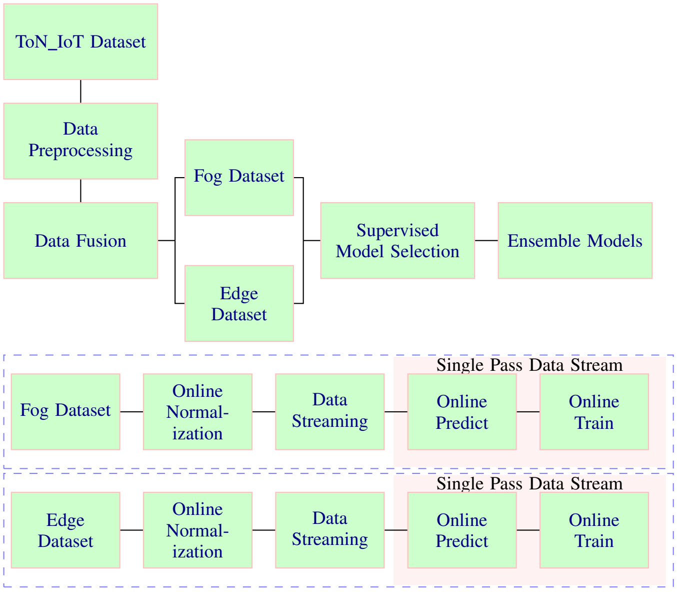

Fig. 2. Data preprocessing flow (up) and online processing of intrusion detection (down).

As illustrated in Fig. 2, The data preprocessing phase begins with dataset preparation and partitioning according to the architectural layers. A representative dataset sample is then utilized to determine the most suitable model for the OCL process. We selected three models that maintain an optimal balance between predictive performance and execution runtime, ensuring the IDS can operate in real-time. Models that offer superior accuracy at the expense of significantly higher latency are considered impractical for the real-time requirements of industrial environments.

To mitigate the catastrophic forgetting, we adopt replay-based strategy, by storing a subset of the training data into a small fixed-memory buffer using reservoir sampling technique [11]. Samples from buffer are then periodically replayed to the model to regain past learned knowledge. To address the problem of severe class imbalance, we designed a class-based partitioning memory buffer management system that segregates data according to their respective classes. Every class has its own an independent reservoir buffer. This schema guarantees that every class—no matter how micro its proportion is in the global stream—gets adequate representation in buffer. Naive reservoir sampling relies purely on random uniform probability to accept incoming data. But in a critical network, we cannot risk having a highly informative historical sample randomly overwritten by a new redundant data. To eliminate this downside, we enhanced the standard reservoir with a deterministic selection criteria by utilizing cosine similarity, ensuring only the most distinct and informative samples are retained. This approach guarantees that a subset of instances from past trained classes are retained and replayed uniformly, thereby preserving critical information regarding minority classes despite their low frequency of occurrence in the global population.

TABLE I
PRELIMINARY EXPERIMENT ON MODEL SELECTION

Classifier Models	Accuracy	Kappa	F1 Score	Latency
NaiveBayes	77.11	0	0	1
DynamicWeightedMajority	97.61	93.25	96.63	2
HoeffdingTree	94.58	84.66	92.33	2
HoeffdingAdaptiveTree	93.88	83.41	91.83	2
EFDT	93.29	80.75	90.38	2
OzaBoost	97.18	92.02	96.01	3
KNN	93.68	82.05	91.03	7
OnlineBagging	96.39	89.79	94.89	9
OnlineSmoothBoost	97.17	91.96	95.98	10
OnlineAdwinBagging	96.95	91.41	95.71	12
LeveragingBagging	97.93	94.20	97.11	21
PassiveAggressiveClassifier	98.85	96.74	98.37	23
SGDClassifier	98.74	96.42	98.21	23
AdaptiveRandomForestClassifier	98.97	97.10	98.56	27
StreamingRandomPatches	98.98	97.14	98.58	43

A. Model Selection

We conducted benchmarking using identical data samples to evaluate and compare the performance of each model. Our model selection was conducted using a benchmarking sample of 45,203 data instances to evaluate and select the three most suitable models to serve as the base learners for our ensemble framework.The complete benchmarking results are presented in Table I. The results shows that Hoeffding Adaptive Tree (HAT), Dynamic Weighted Majority (DWM), and OzaBoost models emerged as the superior choices. These models provide an optimal trade-off between Accuracy, Kappa, and F1-score, while maintaining exceptionally low computationally latency, making them suitable for real-time industrial deployment.

B. Dataset

The ToN_IoT dataset [13] was developed by researchers at the University of New South Wales (UNSW) Canberra. It provides a comprehensive representation of data across the Edge (IoT and telemetry sensors), Fog (network traffic), and Cloud layers. The architecture of the ToN_IoT, coupled with its comprehensive data availability across each hierarchical level, serves as the primary justification for selecting this dataset to design our proposed IDS.

It effectively simulates heterogeneous attacks across various target devices, to address the limitations of legacy datasets by providing a realistic representation of the interdependencies between Edge, Fog, and Cloud environments found in modern industrial ecosystems [14]. In real-world, specific attacks are often localized and directed at particular components rather than occurring as indiscriminate, simultaneous strikes across the entire network infrastructure. Furthermore, it provides distinct ground-truth labels for each targeted component, enabling a granular evaluation of the system's detection capabilities. It also exhibits a significant class imbalance, reflecting real-world cybersecurity environments where attack signatures are substantially less frequent than normal operational traffic [16].

C. Evaluation

In the processing of highly imbalanced datasets, standard metrics such as Accuracy, and F1-score often fail to provide a reliable performance indicator [15]. To address this limitation, alternative metrics are proposed, notably the Matthews Correlation Coefficient (MCC).

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (1)$$

The primary advantage of MCC over traditional metrics lies in its requirement for high-quality results across all four categories of the confusion matrix: True Positives (TP), False Negatives (FN), True Negatives (TN), and False Positives (FP). Specifically, in the context of imbalanced datasets, MCC provides a superior assessment of model reliability, whereas the F1-score and Accuracy may fail to accurately reflect the ratio between positive and negative class elements [15].

D. Experiment Limitations

At the Fog layer, the architecture initially considered three distinct components: Ubuntu Linux, Windows 7, and Windows 10. Preliminary data profiling revealed that the Windows 7 and Windows 10 datasets were insufficient for representing the attack patterns observed at the Edge. Consequently, due to this temporal and volumetric misalignment, the Windows 7 and Windows 10 were excluded from the Fog-IDS dataset to ensure model integrity and evaluation consistency.

Furthermore, both the IDSs were intentionally designed to detect threats exclusively at the Edge layer. Although the ToN_IoT provides distinct data representing specific threats at the Fog layer, this study restricts its evaluation to the Edge layer to ensure early detection of potential threat and taking required counter-measures before it propagate further.

V. Experimentation Results

A. Experimental Settings

Various frameworks have been developed to streamline the deployment of OCL systems. As synthesized by Gomes et al., prominent libraries include MOA and its high-performance wrapper CapyMOA[1] [9], River, and Scikit-Multiflow [10]. Empirical benchmarks have identified MOA and CapyMOA as superior choice for OCL applications, demonstrating significant performance advantages over other alternatives [12].

The experiments were implemented using a legacy-tier laptop (HP EliteBook 820 G2, i7-5600U with 8G of RAM, originally released in 2015) running a Debian operating system to mimic the operational environments of legacy OT networks and was utilized as the primary hardware platform to perform all prequential evaluations, latency benchmarks, and computational resource measurements. The implementation was based on CapyMOA 0.12.0, Python 3.12, NumPy 2.3, and Pandas.

[1]https://capymoa.org/

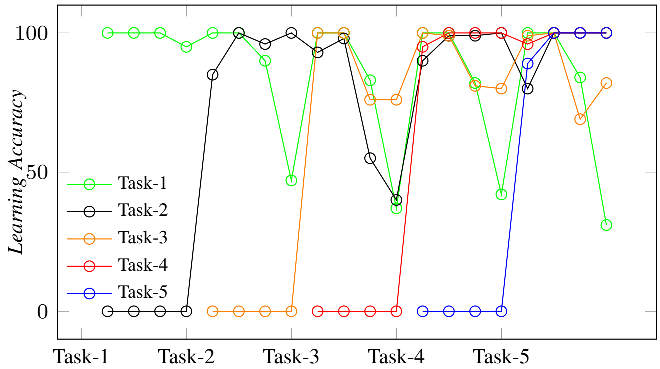

Fig. 3. Plasticity model.

B. Dataset Preparation

Data preprocessing at the Edge layer, six distinct IoT devices are integrated using raw data fusion techniques to construct device-specific datasets. Each device's telemetry and its corresponding ground-truth labels are merged with the broader dataset, resulting in a consistent feature set of 15 variables. However, number of instances (data points) is non-uniform, as it is determined by the specific distribution of ground-truth labels for each individual device.

Each dataset is subsequently transformed into an online data stream using the CapyMOA stream framework. A single instance represents a telemetry data snapshot at a specific point in time. Within the OCL paradigm, the model processes each instance only once; as soon as a new instance arrives, the previous data point is discarded and becomes inaccessible to the model.

C. Plasticity-Stability Evaluation

Plasticity represents the model's capacity to acquire new information upon the arrival of a new task, whereas stability defines the ability to retain historical knowledge after training on new task. In this study, plasticity is measured using Learning Accuracy (LA), denoted as acc_j^i, where the LA of $Task - j$ represents the accuracy of $Task - j$ after the model has been sequentially trained from $Task - 1$ to $Task - i$ (Fig. 3).

Conversely, stability is evaluated through the Forgetting Measure (FM), denoted as fm_j^k. The FM of $Task - j$ is evaluated by measuring the decline in accuracy for $Task - j$ after the model is trained on $Task - i$, relative to the highest accuracy achieved by the model prior to the training of $Task - k$.

$$LA_j = Acc_j^i \quad (2)$$

$$FM_j^k = max(Acc_j^i - Acc_j^k) \quad (3)$$

Plasticity and stability are evaluated across five distinct tasks, each consisting of three unique classes. Fig. 3 illustrates plasticity metric of all five tasks. For all experiments, model

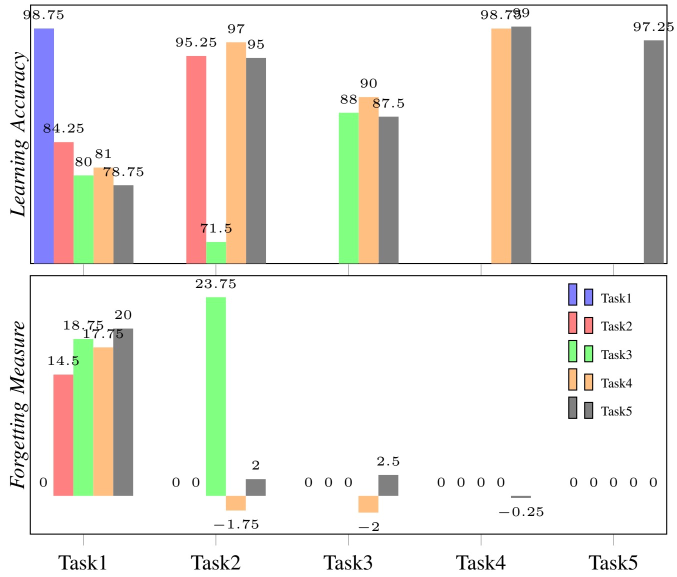

Fig. 4. Plasticity-Stability Dilemma.

demonstrated high plasticity on new task. For instance, LA of $Task-1$ remain high after $Task-2$ to $Task-5$ were introduced. This results shows that model can achieve decent performance in evaluating new tasks.

In terms of stability, as presented in Fig. 4, the model demonstrated a high capability of retaining old knowledge, as evidenced by the marginally lower FM rates. The highest FM rate was recorded for $Task-3$ after the model underwent training on $Task-2$ (23.75%), whereas the remaining tasks demonstrated minor FM rates. The global average FM is 6.05%, calculated using the average of individual FM across all tasks. During the period of increased FM, the achieved LA exhibited a downward trend. Furthermore, the figure illustrates the classic stability-plasticity dilemma, wherein a high LA (indicating high plasticity) was achieved at the expense of stability, as reflected by an increased forgetting measure.

This results demonstrated the effectiveness of the experience replay mechanism in preserving information from historical tasks. Furthermore, the LA for newly introduced tasks was successfully sustained, even after the model was sequentially trained across multiple tasks.

D. Prequential Evaluation Results

The experimental phase involves a comprehensive series of prequential evaluation to examine the performance of both Edge-IDS and Fog-IDS in identifying threat classes across each edge devices. To summarize the IDS performance for each specific threat category, the confusion matrices from all experiments sharing the same threat class are consolidated. This consolidation of confusion matrices across device-specific experiments constitutes the collective performance metric of the IDS for each respective threat class. All experimentation results presented in Table II (threat-wise) and Table III (device-wise), green highlights denote the respective IDS layer exhibiting the superior performance metric.

Table II reveals that Fog-IDS demonstrates higher performance across 7 out of the 8 evaluated threat categories, except in Backdoor where Edge-IDS yields a minor performance advantage. The Scanning class represents the lowest performance tier, yielding moderate MCC of 45.99 and 58.16 for the Edge-IDS and Fog-IDS, respectively.

For Device-wise, as shown in Table III, indicates that the model can achieve optimal performance across all devices, as indicated by high MCC baselines. Intrusion targeting the Garage Door, Motion Light, and Weather subsystems are more optimally classified under Fog-IDS, highlighting the importance of multi-node data consolidation for ICS networks. Conversely, intrusion for the remaining device endpoints yields superior performance at the Edge-IDS, indicating that localized stream analysis retains an operational advantage for highly targeted node-specific intrusions. Overall, the model achieved Accuracy of 99.58, Kappa of 98.21, F1-score of 98.46, and MCC of 98.21 at Fog-IDS, while securing Accuracy of 99.45, Kappa of 97.68, F1-score of 97.99, and MCC of 97.68 at Edge-IDS.

In terms of computational resource efficiency, the model utilized approximately 480 MB to 560 MB of RAM and 19% to 25% of the CPU load throughout the evaluation phase. For latency measurements, the model was capable of processing approximately 1,763 iterations per second (It/s) during prequential phase, 5,648 It/s during the train-only phase, and 6,921 It/s during the test-only phase.

The model demonstrated highly reliable performance in identifying Injection, Backdoor, Normal, Password, DDoS, Ransomware, and XSS tasks. Conversely, the classification performance for the Scanning task was categorized as moderate, as indicated by MCC values below 0.7. This performance

TABLE II
THREAT-WISE PREQUENTIAL EVALUATION RESULTS

	Accuracy		F1 score		Kappa		MCC	
Threat	**Edge**	**Fog**	**Edge**	**Fog**	**Edge**	**Fog**	**Edge**	**Fog**
Backdoor	99.76	99.76	95.51	95.41	95.39	95.28	95.43	95.34
DDoS	99.46	99.95	86.44	98.58	86.17	98.56	86.56	98.56
Injection	98.35	98.64	98.97	99.15	94.86	95.79	94.9	95.84
Normal	99.35	99.35	96.72	96.79	96.36	96.43	96.36	96.46
Password	99.45	99.59	94.59	96.03	94.3	95.82	94.3	95.84
Ransomware	99.9	99.91	93.12	93.69	93.07	93.64	93.09	93.69
Scanning	99.66	99.8	44.17	57.97	44.02	57.87	45.99	58.16
XSS	99.79	99.84	66.07	71.27	65.97	71.19	68.04	72.16

TABLE III
DEVICE-WISE PREQUENTIAL EVALUATION RESULTS

	Accuracy		F1 score		Kappa		MCC	
Device	**Edge**	**Fog**	**Edge**	**Fog**	**Edge**	**Fog**	**Edge**	**Fog**
Fridge	99.73	99.64	99.05	98.72	98.9	98.51	98.9	98.51
Garage Door	98.45	99.31	93.79	97.23	92.9	96.84	92.9	96.84
Modbus	99.35	99.22	98.04	97.66	97.65	97.19	97.65	97.19
Motion light	98.83	99.26	95.34	97.04	94.67	96.62	94.67	96.62
Thermostat	99.75	99.72	99.11	99.03	98.96	98.87	98.96	98.87
Weather	99.68	99.79	98.7	99.16	98.52	99.04	98.52	99.04

TABLE IV
DOMINANT MODELS FOR EACH THREAT SCENARIO

Threat	Fog-IDS	Edge-IDS
Backdoor	HoeffdingAdaptiveTree	OzaBoost
DDoS	HoeffdingAdaptiveTree	DynamicWeightedMajority
Injection	HoeffdingAdaptiveTree	OzaBoost
Normal	HoeffdingAdaptiveTree	OzaBoost
Password	HoeffdingAdaptiveTree	OzaBoost
Ransomware	HoeffdingAdaptiveTree	DynamicWeightedMajority
Scanning	OzaBoost	OzaBoost
XSS	OzaBoost	OzaBoost

TABLE V
BASELINE COMPARISONS

Ref, Year	Methods	F1 score	Accuracy
[18], 2024	CNN-GRU	99.10	99.13
[19], 2025	CapsNets, Attention Augmented RNN, QIPSO, ANFIS	98.65	98.83
[20], 2025	Ensemble (RF - GBM -SVM)	97.10	98.70
Ours	Class-based Reservoir with Similarity Selection	98.46	99.58

trend was consistently observed across both the Edge-IDS and Fog-IDS architectures.

Furthermore, it is essential to analyze the performance of the individual models constituting the IDS. A distinct phenomenon is observed wherein no single model demonstrates universal dominance across all categories. Instead, specific threat classes are more effectively identified by particular models as summarized in Table IV.

For instance, the HAT outperformed others in detecting the Backdoor, DDoS, Injection, and Password at Fog-IDS. In contrast, the DWM was most effective for DDoS and Ransomware detection at Edge-IDS. Finally, the OzaBoost model proved superior for identifying Scanning and XSS at both IDS. Observations of individual model performance within both Edge-IDS and Fog-IDS ensembles indicate a clear specialization; each threat category and dataset type requires a specific model architecture to achieve optimal detection performance.

A baseline comparison was also conducted against several recent studies related to the ToN_IoT dataset, as presented in Table V. However, because the cited works did not explicitly report their MCC scores, the comparative evaluation was strictly limited to Accuracy and F1 score values.

VI. CONLUSION

The implementation of OCL-based intrusion detection proves that localized, real-time threat detection is highly effective for ICS. By utilizing Class-based reservoir with Similarity selection, our ensemble model maintains high performance (Accuracy of 99.58, Kappa of 98.21, F1-score of 98.46, and MCC of 98.21 at Fog-IDS; and Accuracy of 99.45, Kappa of 97.68, F1-score of 97.99, and MCC of 97.68 at Edge-IDS) while preserving data privacy and operating within the resource constraints of Edge and Fog layers.

REFERENCES

[1] Abdullah Alsaedi et al. "TON IoT Telemetry Dataset: A New Generation Dataset of IoT and IIoT for Data-Driven Intrusion Detection Systems". In: IEEE Access 8 (2020), DOI: 10.1109/ACCESS.2020.3022862.

[2] Tung-Anh Nguyen et al. "Federated PCA on Grassmann Manifold for IoT Anomaly Detection". In: IEEE/ACM Transactions on Networking 32.5 (2024), DOI: 10.1109/TNET.2024.3423780.

[3] Motong Sun et al. "Intrusion detection system based on in-depth understandings of industrial control logic". In: IEEE Transactions on Industrial Informatics 19.3 (2022), DOI: 10.1109/TII.2022.3200363

[4] S. Zhang, Y. Xu and X. Xie, "Universal Adversarial Perturbations Against Machine-Learning-Based Intrusion Detection Systems in Industrial Internet of Things," in IEEE Internet of Things Journal, doi: 10.1109/JIOT.2024.3465549.

[5] Eirini Anthi et al. "Adversarial attacks on machine learning cybersecurity defences in industrial control systems". In: Journal of Information Security and Applications 58 (2021), DOI: 10.1016/j.jisa.2020.102717.

[6] Ammar Alazab et al. "Enhancing privacy-preserving intrusion detection through federated learning". In: Electronics 12.16 (2023), DOI: 10.3390/electronics12163382.

[7] Liyuan Wang et al. "A comprehensive survey of continual learning: Theory, method and application". In: IEEE transactions on pattern analysis and machine intelligence 46.8 (2024), DOI: 10.1109/TPAMI.2024.3367329"

[8] Nicholas Jeffrey, Qing Tan, dan Jose R. Villar. "A Review of Anomaly Detection Strategies to Detect Threats to Cyber-Physical Systems". In: Electronics (Switzerland) 12.15 (2023). DOI: 10.3390/electronics12153283.

[9] Albert Bifet et al. "MOA: Massive online analysis, a framework for stream classification and clustering". In: Proceedings of the first workshop on applications of pattern analysis. PMLR. 2010, URL: https://proceedings.mlr.press/v11/bifet10a.html

[10] Jacob Montiel et al. "River: machine learning for streaming data in python". In: Journal of Machine Learning Research 22.110 (2021), DOI: 10.48550/arXiv.2012.04740.

[11] Jeffrey S Vitter. "Random sampling with a reservoir". In: ACM Transactions on Mathematical Software (TOMS) 11.1 (1985).

[12] Heitor Murilo Gomes et al. "CapyMOA: efficient machine learning for data streams in python". DOI: 10.48550/arXiv.2502.07432.

[13] M. Booij et al. "ToN IoT: The Role of Heterogeneity and the Need for Standardization of Features and Attack Types in IoT Network Intrusion Data Sets". In: IEEE Internet of Things Journal 9.1 (2022), DOI: 10.1109/JIOT.2021.3085194.

[14] Nour Moustafa. "New generations of internet of things datasets for cybersecurity applications based machine learning: TON IoT datasets". In: Proceedings of the eResearch Australasia Conference, Brisbane, Australia. 2019, DOI: 10.26190/5d7ac9bfe8487.

[15] Davide Chicco dan Giuseppe Jurman. "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation". In: BMC genomics 21.1 (2020), DOI: 10.1186/s12864-019-6413-7.

[16] Harwahyu, R., Ndolu, F. H. E., & Overbeek, M. V. (2024). Three layer hybrid learning to improve intrusion detection system performance. International Journal of Electrical and Computer Engineering, 14(2), 1691-1699. 10.11591/ijece.v14i2.pp1691-1699.

[17] Daulay, A., Ramli, K., Sudiana, D., Harwahyu, R., Hidayat, T., & Fachrurrozi, N. R. (2025). A Novel Hybrid Model for High-Accuracy Malware Detection in The Internet of Medical Things (IoMT) Environment. DOI:10.31436/iiumej.v26i3.3746

[18] M. Maaz, G. Ahmed, A. Sami Al-Shamayleh, A. Akhunzada, S. Siddiqui and A. Hussein Al-Ghushami, "Empowering IoT Resilience: Hybrid Deep Learning Techniques for Enhanced Security," in IEEE Access, vol. 12, 2024, doi: 10.1109/ACCESS.2024.3482005.

[19] G. Logeswari, J. Deepika Roselind, K. Tamilarasi and V. Nivethitha, "A Comprehensive Approach to Intrusion Detection in IoT Environments Using Hybrid Feature Selection and Multi-Stage Classification Techniques," in IEEE Access, vol. 13, 2025, doi: 10.1109/ACCESS.2025.3532895.

[20] M. Al-Hubaishi and M. Hachana, "Enhanced Intrusion Detection for IoT Networks Using Machine Learning Approach" 2025 9th International Symposium on Innovative Approaches in Smart Technologies (ISAS), Gaziantep, Turkiye, 2025, doi: 10.1109/ISAS66241.2025.11101771.